engineering 4 us all

Student Handout

Name: ______________________________ **Date:** _________________ **Class:** _______________

MyDesign Portfolio

Element L Initial Template AT - 6 points

Element L: Presentation of Designer's Recommendations

Content of Recommendations *(L1, L2)*

- What could be done next to go further with this project?

Taking into account the concerns raised about the mold-resistant coating, we believe using a more plant-safe, biodegradable alternative would improve environmental safety because it reduces the risk of chemical damage to plants. Based on expert advice about water distribution, adding a rotating or adjustable nozzle system would improve even watering because it can adapt to different plant layouts. From structural feedback, reinforcing the pipe with modular metal brackets would improve durability because it prevents sagging and makes maintenance easier. Also, we recommend testing in different environmental conditions (temperature, humidity) to collect more comprehensive data on performance.

- What would you keep if you did the project again?

If we did this project again, we would keep the plant spacing and the overall layout of the watering system, as the expert confirmed it allowed good distribution and strength. We would also retain the idea of using a mold-resistant coating, but ensure it is safer for plants. The general structure of the pipe system with support underneath worked well and was easy to assemble. Our testing for water flow and stability was successful and provided useful data. We would also keep the feedback review process, as it helped guide better iterations.

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- What would you change in your current prototype or testing plan?

We would change the method of applying the mold-resistant coating by using a removable cover or enclosed space to protect plants during application. To improve water distribution, we would upgrade to a multi-nozzle misting system with adjustable angles for even coverage. We would design a quick-connect pipe system for easy maintenance, using a twist-and-lock feature based on stakeholder feedback. For testing, we would add moisture sensors to collect objective data on soil hydration over time and include longer test periods. These changes would make the prototype more functional and easier to test in real-life conditions.

- What would have made this project easier for you to complete?

  - Having easier access to testing tools like moisture sensors and flow meters.
  - Getting earlier feedback from field experts during the initial design phase.
  - More time allocated for building and iterating the prototype.
  - A clearer timeline or checklist to help manage tasks better.
  - Extra materials on hand for quick fixes or changes during prototyping.